Panskura Banamali College (Autonomous)

B.Sc.(H) 4th Semester Examination -2021 (Online Mode)

Subject: Computer Science **Paper: CC10T**

Paper Name: Database Management Systems

F.M.: 40 **Time : 2 Hours**

Answer any four : **4x10=40**

1. Define with example DBMS , DBA , DML , DCL , DDL . 5×02=10

2. What is schema ? What is attribute ? What is weak entity ? Write difference between primary key and candidate key . Define foreign key . 4× 2.5=10

3. Draw a proper E-R diagram of a online railway reservation system . Map this E-R diagram to relational model . 6+4=10

4. a) What is an integrity constraint? State different types of integrity constraint.

   b) What is RDBMS? What is FDs? State with example.

   c) State three level architecture of DBMS. 3+4+3=10

5. a) What is data model? State Hierarchical and network data model with advantage and disadvantage.

b) why normalization is necessary in RDBMS?.

c) "All BCNF is a 3NF but all 3 NF is not BCNF"------justify . (3+3)+2+2=10

6.short note (any four)

a) foreign key b) normalization c)Mapping d) natural join e)aggregation

f) lossless join decomposition. 2.5+2.5+2.5+2.5=10

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